

Garrett Fincke

Pittsburgh, PA | garrettfinke@gmail.com | fincke.dev | linkedin.com/in/garrett-fincke | github.com/ggfincke

Education

University of Pittsburgh

Master of Science in Computer Science

Aug 2026 – Expected May 2028

Pittsburgh, PA

The Pennsylvania State University

Bachelor of Science in Computer Science

Aug 2021 – Dec 2024

University Park, PA

Experience

Westinghouse Electric Company

Software Developer Intern, GIC Tools and Applications

May 2026 – Aug 2026

Cranberry Township, PA

- Building a Python/FastAPI service over an isolated Oracle Document Index, exposing baseline, release, and configuration-item data to engineering and AI workflows
- Implementing Active Directory basic auth, project-level Oracle authorization, API keys, rate limiting, and concurrent JSON access for restricted-data environments
- Integrating with an internal agentic AI platform (Databricks lakehouse) as a source-grounded retrieval tool for LLM-assisted I&C engineering workflows

Scale AI

Software Engineer (contract)

May 2024 – Jul 2025

Remote

- Built Python evaluation harnesses for LLM-generated code, compiling and executing submissions in sandboxed Docker environments and scoring correctness against automated tests
- Automated linting, test orchestration, and metrics pipelines to track pass@k, syntax/style errors, and failure modes across code-generation tasks
- Created schema-versioned data pipelines using Pydantic, JSON Schema, and deterministic sampling, replacing ad hoc manual review with reproducible scripted validation

Projects

SwimMate v2 | *Python, Django REST, PostgreSQL/pgvector, OpenAI, Swift, SwiftUI, iOS/watchOS*

Full-stack swimming platform with native iOS/watchOS apps and AI-generated workouts

- Built Django REST API with JWT auth, pgvector semantic search, and RAG-based workout generation
- Implemented constraint-aware generation with stroke/difficulty/intensity reranking, redundancy penalties, template/set support, and multi-level pace resolution
- Developed SwiftUI iOS/watchOS clients with HealthKit workout capture, Apple Watch metrics, Swift Charts analytics, WatchConnectivity sync, and goal-based workout tracking

MDX Preview for VS Code | *TypeScript, React, MDX, VS Code Extension API, Vite*

Published VS Code/Open VSX extension for live MDX previews with 3,500+ installs

- Built 6-package monorepo with esbuild extension-host builds, Vite webviews, and shared runtime/compiler tooling published as a reusable npm library
- Implemented framework-aware MDX resolution across 5 frameworks; shipped 19 commands, 30 settings, themes, and automated quality gates

Minecart | *TypeScript, Node.js, Discord.js, Docker, PostgreSQL, RCON, AWS EC2/SSM/CloudWatch*

Multi-tenant platform for managing Minecraft servers across local Docker and AWS EC2

- Built Discord dashboards for server operations, logs, backups, worlds, configuration, and performance across Docker/RCON and EC2/SSM providers
- Implemented per-guild isolation, RBAC, audit logging, quota enforcement, and blue-green staging

Technical Skills

Languages: Python, TypeScript/JavaScript, Swift, Java, SQL, C/C++

Backend & Data: Django, FastAPI, Node.js, PostgreSQL, Oracle, Redis, Celery, Pydantic, REST APIs

AI/LLM: RAG, embeddings/vector search, agentic AI, MCP, Ollama, vLLM, OpenAI/Anthropic APIs

Frontend & Mobile: React, Next.js, SwiftUI, UIKit, Tailwind CSS

Cloud & DevOps: Docker, Docker Compose, AWS EC2/SSM/CloudWatch, GitHub Actions, CI/CD, Vercel